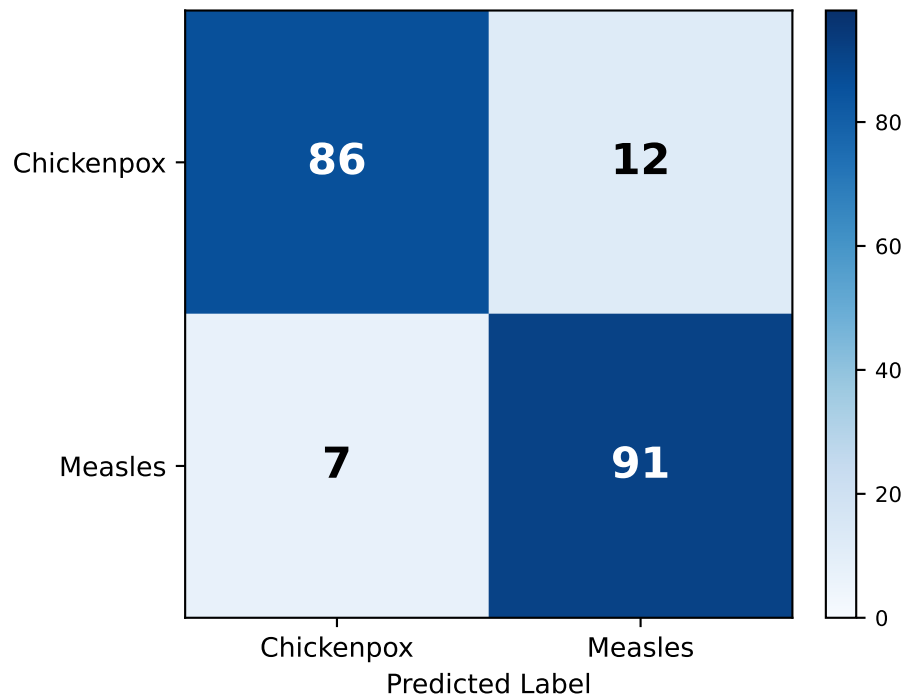
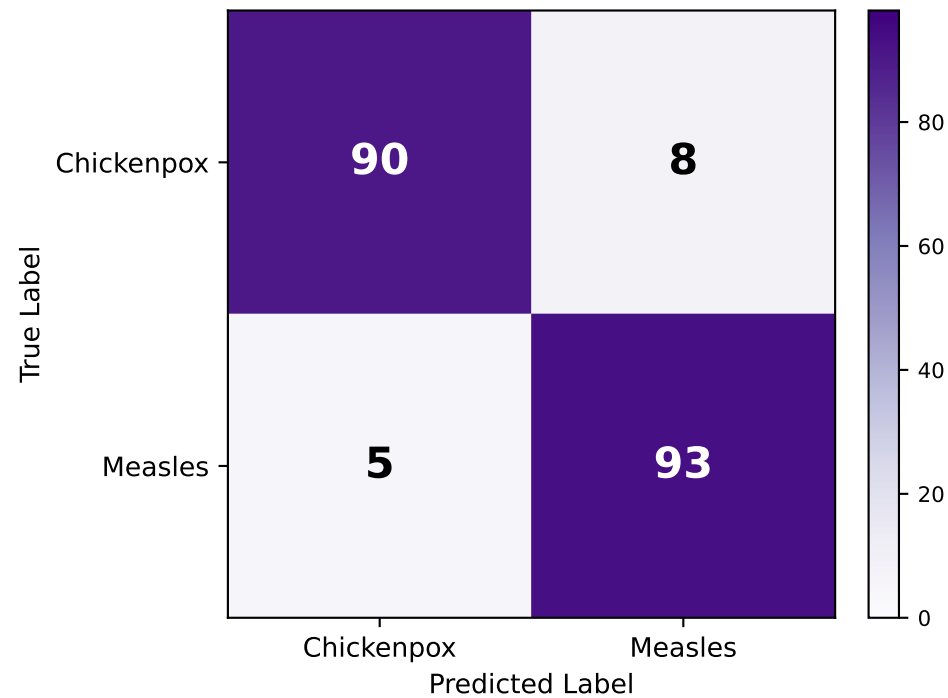


Holdout Confusion Matrices: All Four Experimental Phases (196-Sample Unseen Test Partition)

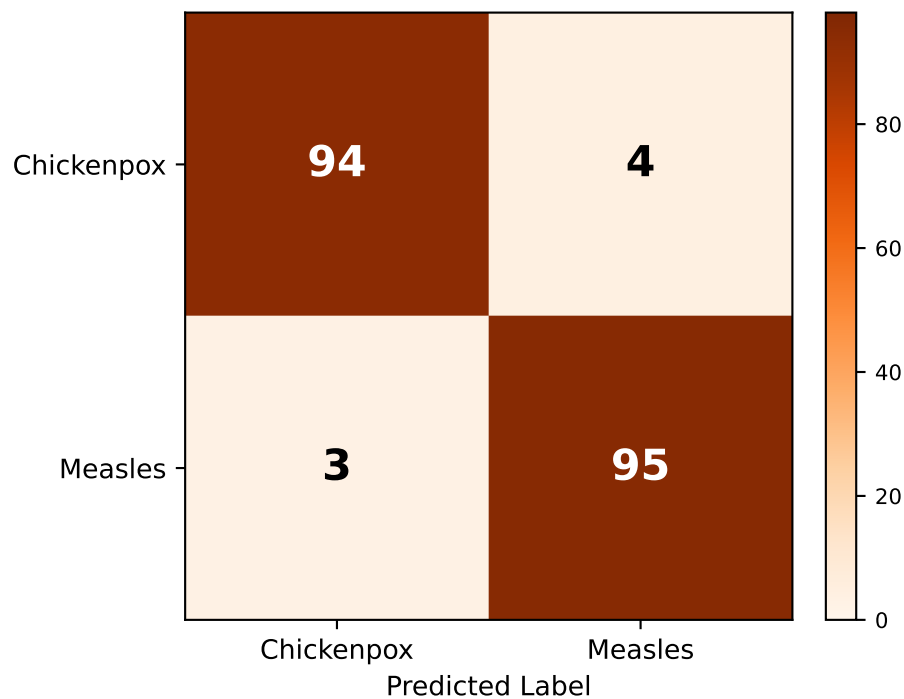
**Phase 1: Random Forest
(Morphological Only) F1 = 0.901**



**Phase 2: EfficientNet-B0
(Frozen Transfer) F1 = 0.934**



**Phase 3: EfficientNet-B0
(Fine-Tuned) F1 = 0.963**



**Phase 4: MorphNN MobileNetV2+RF [Ours]
(Hybrid Fusion) F1 = 0.957**

